

HARP-Select: Validation-Calibrated Routing between Residual Graph Experts

Fixed Graph Split



$$G = (V, E), X; Y_{tr}, Y_{val}$$

FIXED SPLIT

Two Structural Views



self-loop $\hat{A}$



no-self $\tilde{A}$

STRUCTURAL PRIORS

Parallel Residual Polynomial Experts

identical labels, masks, and optimization budget

Ⓜ HARP-GNN



self-loop



basis bank



node gate



a_H
val. score

Ⓛ HARP-Esep



ego-separated



basis bank



ego fusion



a_E
val. score

Residual Polynomial Basis Zoom

H: self-loop family



$$L_k = \hat{A}^k X$$

$$H_k = L_{k-1} - L_k$$

E: no-self family



$$B_k = \tilde{A}^k B_0$$

$[B_0, U]$ fusion

SPECTRAL BASES

Validation-Calibrated Router

validation labels only

a_H

a_E

$$\Delta_{val} = a_E - a_H$$

retain H select E

$$\tau = 1.96 SE$$

Locked Test Evaluation



test score + oracle regret

TEST LABELS NEVER ROUTE